Burak Ömür

Senior AI Engineer

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PROFILE

AI engineer building and running production LLM systems since 2022. Lead architect of an agentic coaching product at Retorio, owning the runtime, model routing, memory, and observability end to end under EU data residency and GDPR constraints.

EXPERIENCE

Retorio GmbH Munich, DE — Mar 2022 – Present

Senior AI Engineer (Working Student until Jun 2024)

- Lead the architecture of **Tori**, an agentic coaching product, as the sole AI engineer on the team, covering runtime, model routing, state and memory, and observability.
- Built the runtime on a self-hosted Agent Protocol server (Aegra) with LangGraph and DeepAgents for long-horizon planning and subagent delegation, adding durable event logging (Redis Streams into PostgreSQL) and OpenTelemetry tracing into Langfuse, making multi-turn sessions replayable and auditable.
- Consolidated all model traffic behind a central LiteLLM proxy on Cloud Run against Vertex AI EU endpoints, adding Redis-backed fleet-wide admission control, isolating real-time and feedback paths, and eliminating roughly 60-second cold starts.
- Designed per-user memory banks and a source-addressed evidence store with bi-temporal provenance (PROV-O style) to support cited feedback and GDPR erasure requests.
- Defined the SLI and measurement approach behind a contractual SLA for an enterprise customer, based on real user traffic and application-layer error signals that surface mid-stream failures Cloud Run metrics miss.
- Before Tori, built FastAPI and gRPC microservices on Docker and GCP, trained in-house face detection and NLP models, and shipped LLM chains with retry and fallback handling.

Tazi.ai Istanbul, TR — Jul 2020 – Sep 2020

Machine Learning Engineer, Intern

- Built a feature selection method based on variational autoencoders in Scala and Deeplearning4j.

SÜTAŞ Süt Ürünleri A.Ş. Istanbul, TR — Jun 2019 – Aug 2019

Software Engineer, Intern

- Built an internal Android application to speed up communication between production teams and management.

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, Scala

LLM & Agents: LangGraph, LangChain, DeepAgents, Agent Protocol (Aegra), MCP / FastMCP, LiteLLM, RAG

Observability: Langfuse, OpenTelemetry, Grafana

Backend & Data: FastAPI, gRPC, Pydantic, Docker, Redis, PostgreSQL / Cloud SQL, PgBouncer, MongoDB

ML & Cloud: PyTorch, TensorFlow, JAX, computer vision; GCP (Cloud Run, Cloud SQL, Vertex AI), Azure, CI/CD

Compliance: GDPR, EU AI Act, EU data residency

RESEARCH

Towards Smarter and Context-Aware Retrieval Augmented Generation with LLMs *M.Sc. Thesis, TUM*

- Built a query taxonomy for RAG failure modes (feature extraction, disambiguation, hallucination) and compared explicit against implicit feature extraction.
- Created AbbrvQA, an AI-generated ambiguity dataset, and proposed a RAG pipeline with disambiguation and validation modules.

Investigating Bias and Discrimination in Movies Using Deep Learning *B.Sc. Thesis, Boğaziçi*

- Large-scale computer vision analysis of the MovieNet dataset using pre-trained architectures in Python.

EDUCATION

Technical University of Munich Munich, DE — Oct 2021 – Jun 2024

M.Sc. Informatics (GPA 1.61)

Boğaziçi University Istanbul, TR — Sep 2016 – Jun 2021

B.Sc. Computer Engineering (GPA 3.75/4.00)

Languages: Turkish (native), English (fluent), German (A2)